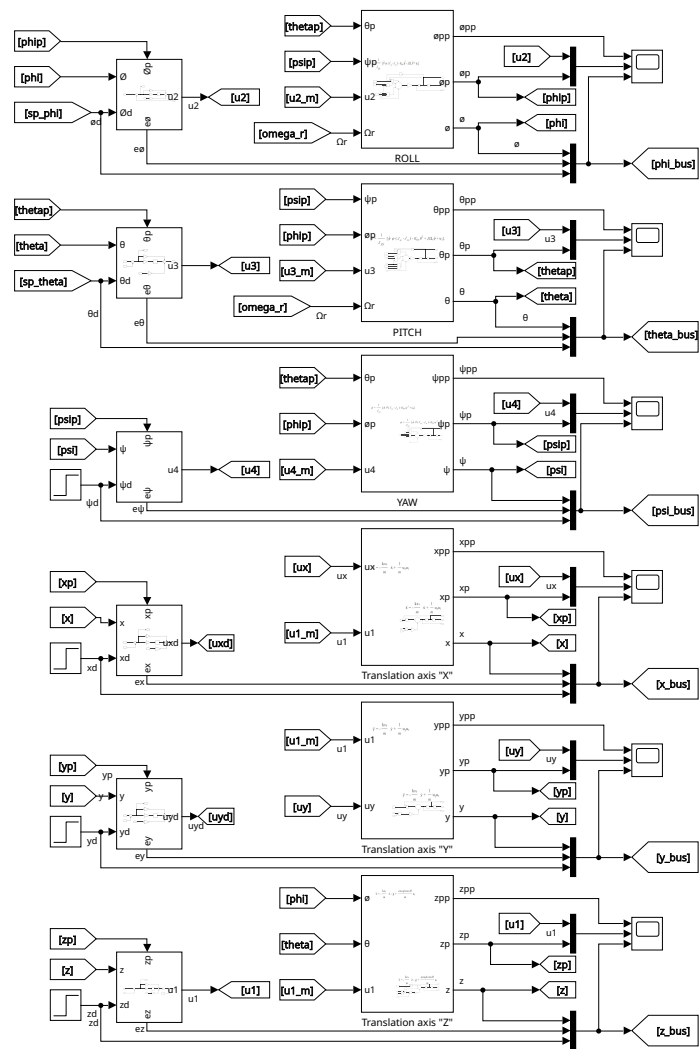
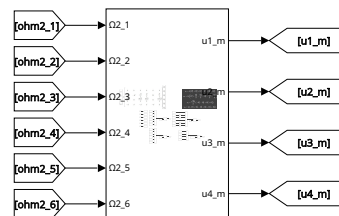
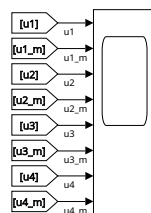
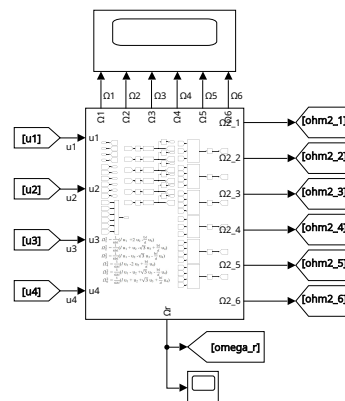
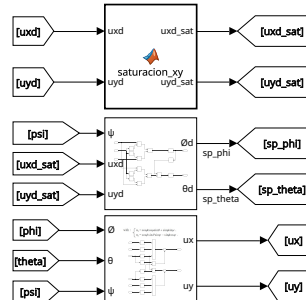
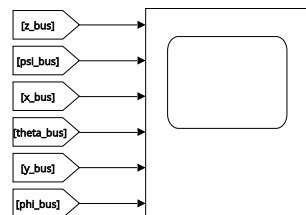




```
function [uxd_sat, uyd_sat] = saturacion_xy(uxd, uyd)
% Saturación vectorial de las referencias horizontales
```

```
% Inclinación máxima permitida
alpha_max = 10*pi/180;
```

```
% Magnitud horizontal máxima
rho_max = sin(alpha_max);
```

```
% Magnitud solicitada por los controladores X-Y
rho = sqrt(uxd^2 + uyd^2);
```

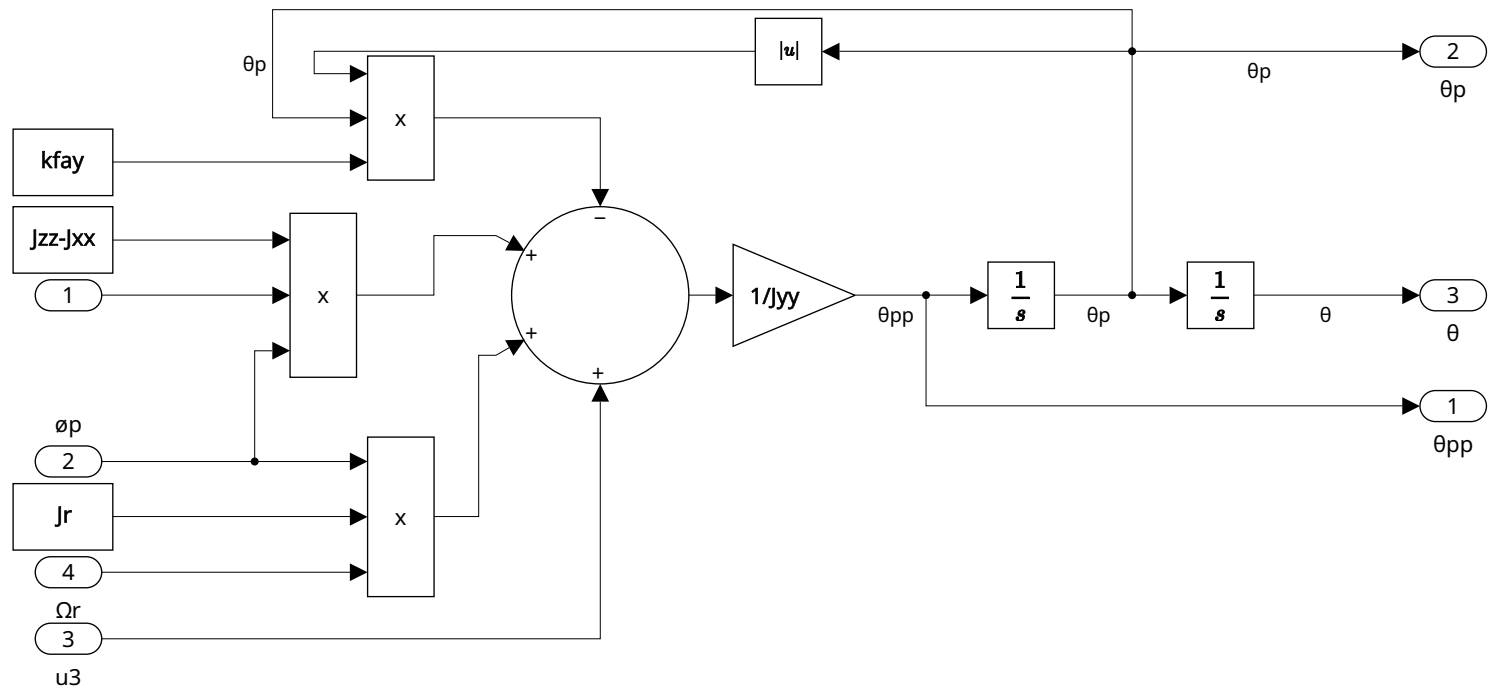
```
% Saturación conservando la dirección
if rho > rho_max
    escala = rho_max / max(rho, 1e-6);
```

```
    uxd_sat = escala * uxd;
    uyd_sat = escala * uyd;
```

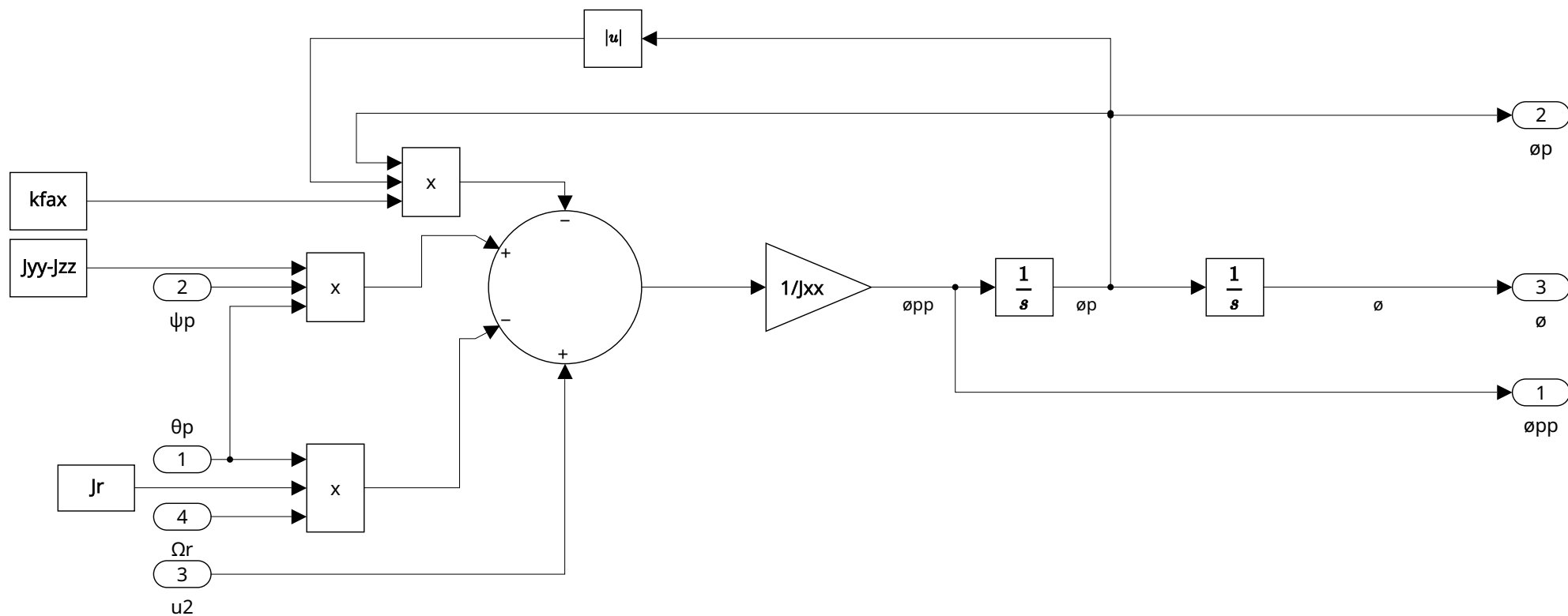
```
else
    uxd_sat = uxd;
    uyd_sat = uyd;
```

```
end
end
```

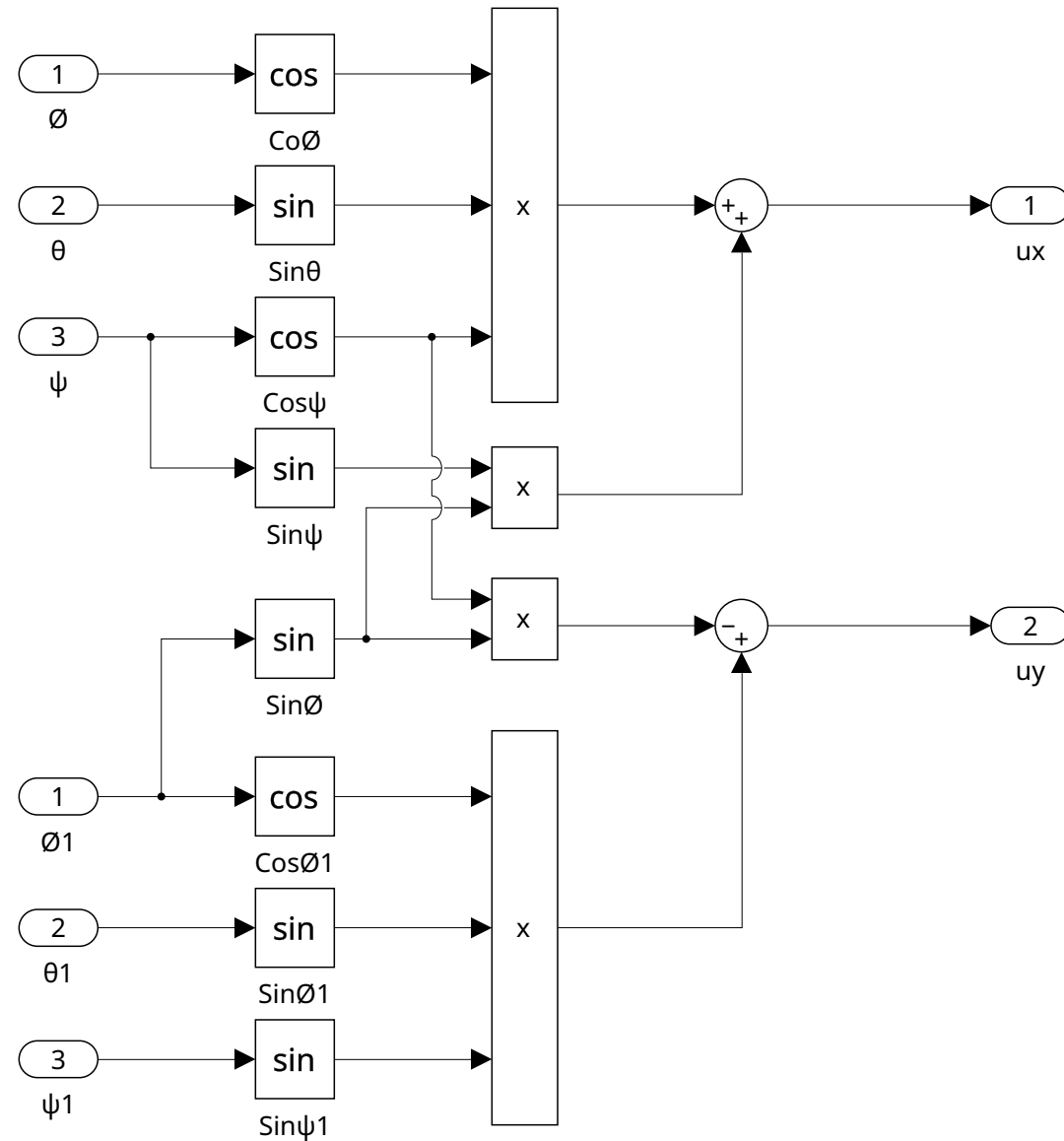
$$\ddot{\theta} = \frac{1}{J_{yy}} [\dot{\phi} \dot{\psi} (J_{zz} - J_{xx}) - K_{fay} \dot{\theta}^2 + J_r \Omega_r \dot{\phi} + u_3].$$

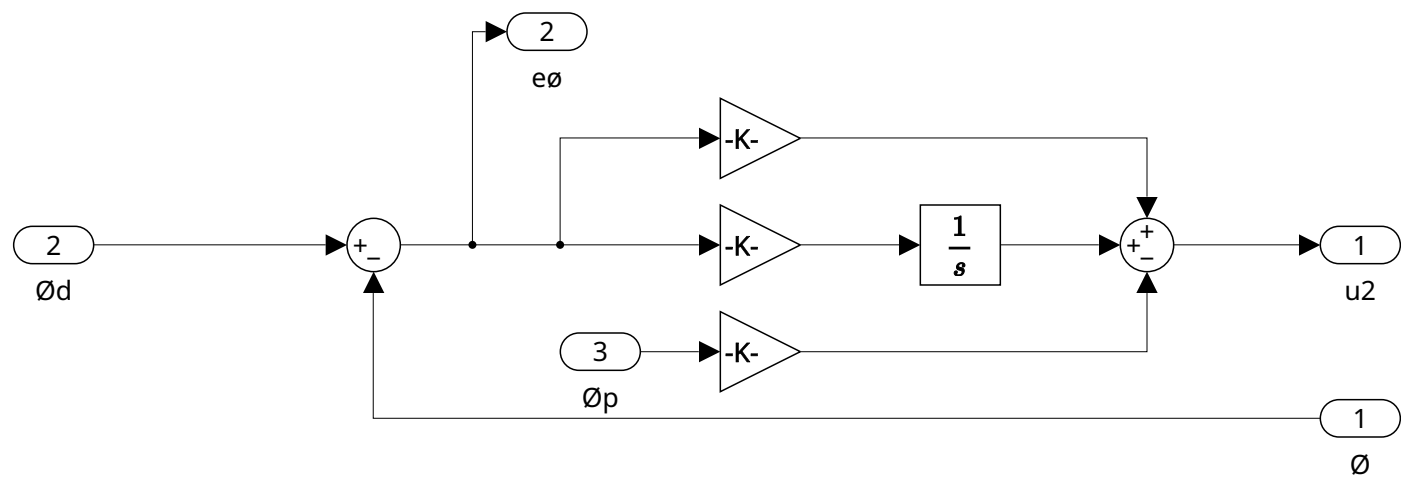


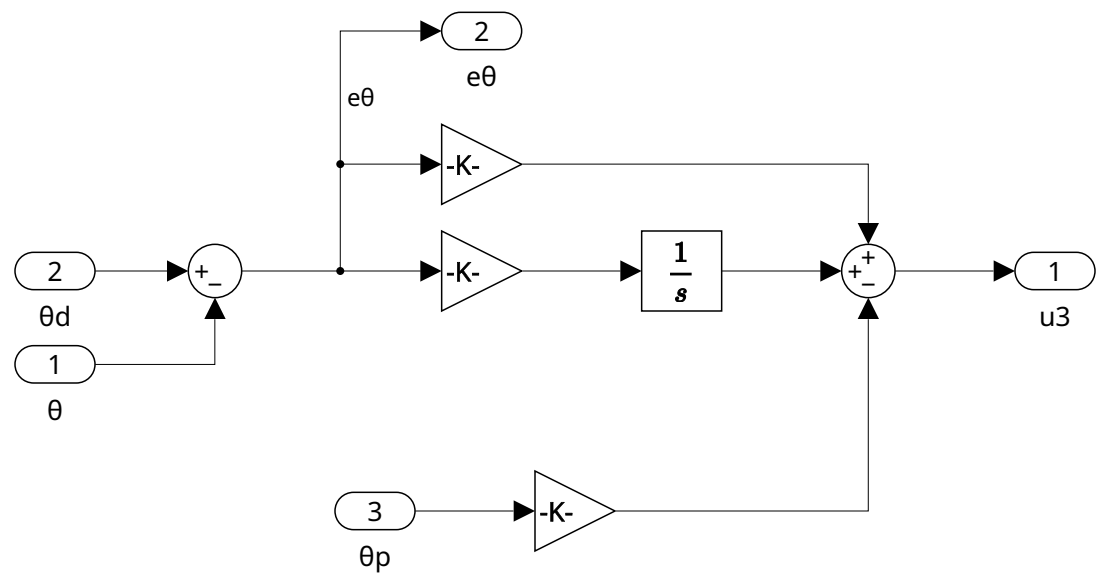
$$\ddot{\phi} = \frac{1}{J_{xx}} [\dot{\theta} \dot{\psi} (J_{yy} - J_{zz}) - K_{fax} \dot{\phi}^2 - J_r \Omega_r \dot{\theta} + u_2].$$

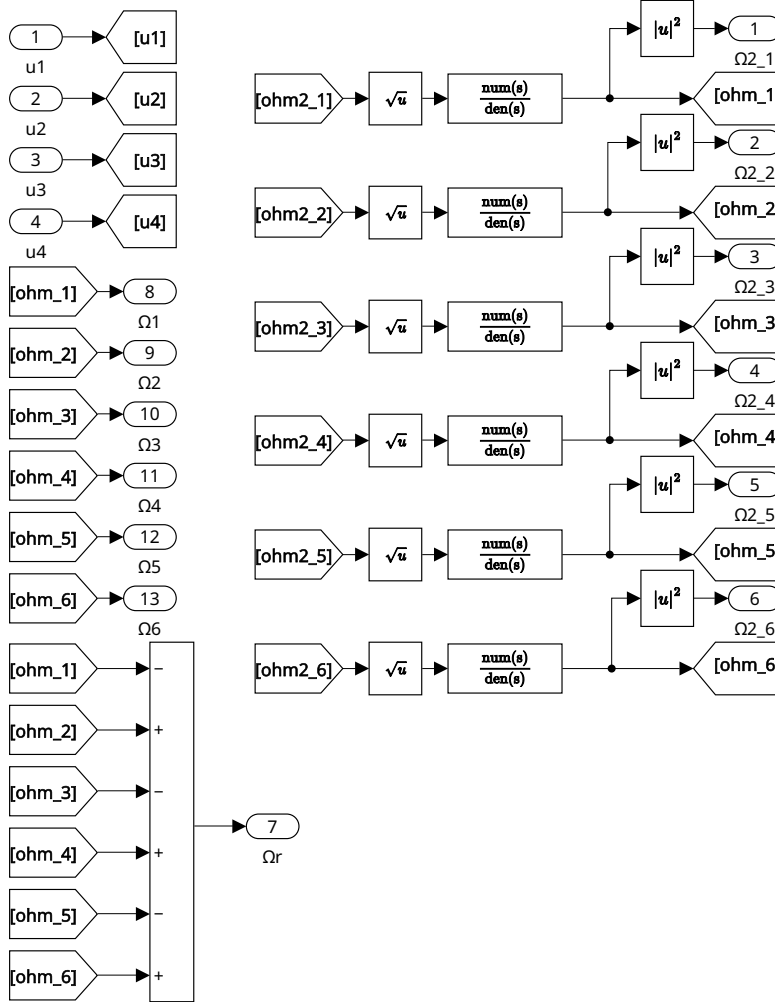


with :
$$\begin{cases} u_x = \cos\phi \cos\psi \sin\theta + \sin\phi \sin\psi \\ u_y = \cos\phi \sin\theta \sin\psi - \sin\phi \cos\psi \end{cases}$$

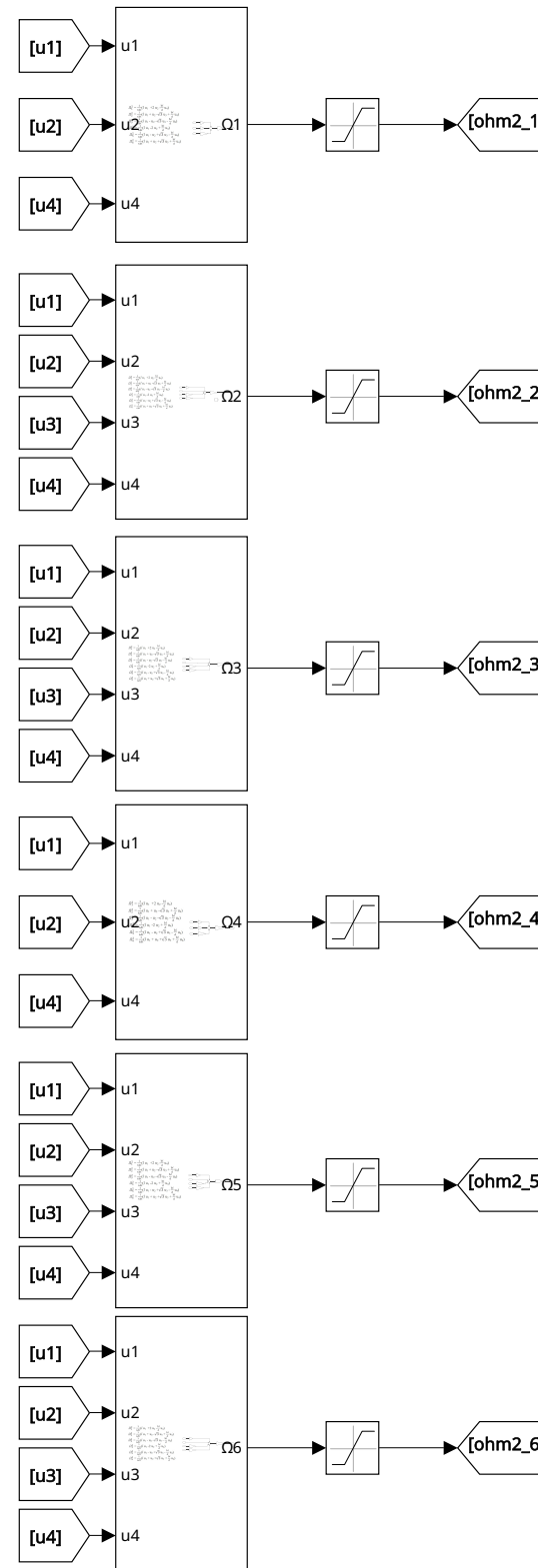








$$\begin{aligned}\Omega_1^2 &= \frac{1}{6bl}(l u_1 + 2 u_2 - \frac{bl}{d} u_4) \\ \Omega_2^2 &= \frac{1}{6bl}(l u_1 + u_2 - \sqrt{3} u_3 + \frac{bl}{d} u_4) \\ \Omega_3^2 &= \frac{1}{6bl}(l u_1 - u_2 - \sqrt{3} u_3 - \frac{bl}{d} u_4) \\ \Omega_4^2 &= \frac{1}{6bl}(l u_1 - 2 u_2 + \frac{bl}{d} u_4) \\ \Omega_5^2 &= \frac{1}{6bl}(l u_1 - u_2 + \sqrt{3} u_3 - \frac{bl}{d} u_4) \\ \Omega_6^2 &= \frac{1}{6bl}(l u_1 + u_2 + \sqrt{3} u_3 + \frac{bl}{d} u_4)\end{aligned}$$



$$\Omega_1^2 = \frac{1}{6bl}(l u_1 + 2 u_2 - \frac{bl}{d} u_4)$$

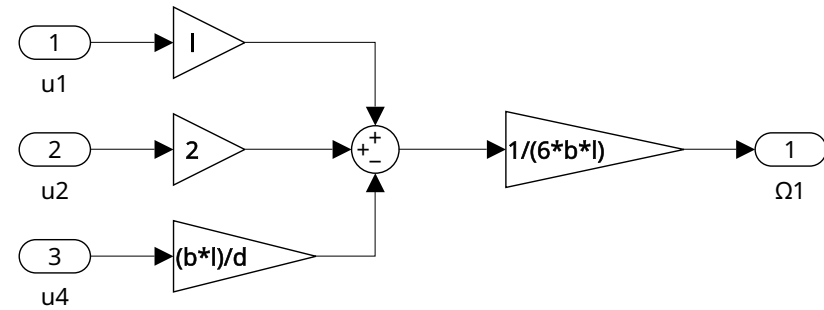
$$\Omega_2^2 = \frac{1}{6bl}(l u_1 + u_2 - \sqrt{3} u_3 + \frac{bl}{d} u_4)$$

$$\Omega_3^2 = \frac{1}{6bl}(l u_1 - u_2 - \sqrt{3} u_3 - \frac{bl}{d} u_4)$$

$$\Omega_4^2 = \frac{1}{6bl}(l u_1 - 2 u_2 + \frac{bl}{d} u_4)$$

$$\Omega_5^2 = \frac{1}{6bl}(l u_1 - u_2 + \sqrt{3} u_3 - \frac{bl}{d} u_4)$$

$$\Omega_6^2 = \frac{1}{6bl}(l u_1 + u_2 + \sqrt{3} u_3 + \frac{bl}{d} u_4)$$



$$\Omega_1^2 = \frac{1}{6bl}(l u_1 + 2 u_2 - \frac{bl}{d} u_4)$$

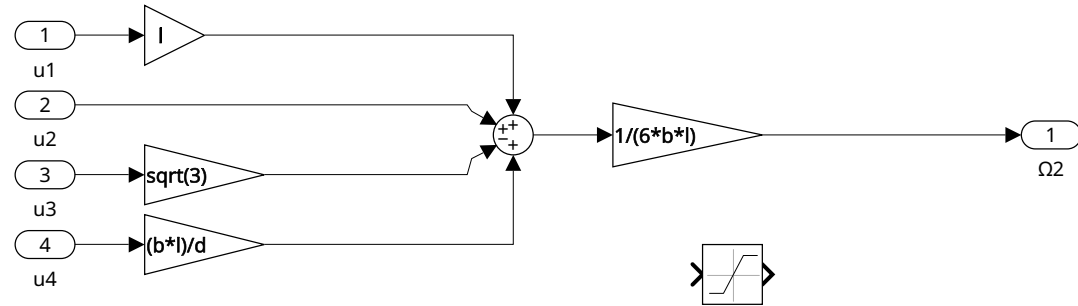
$$\Omega_2^2 = \frac{1}{6bl}(l u_1 + u_2 - \sqrt{3} u_3 + \frac{bl}{d} u_4)$$

$$\Omega_3^2 = \frac{1}{6bl}(l u_1 - u_2 - \sqrt{3} u_3 - \frac{bl}{d} u_4)$$

$$\Omega_4^2 = \frac{1}{6bl}(l u_1 - 2 u_2 + \frac{bl}{d} u_4)$$

$$\Omega_5^2 = \frac{1}{6bl}(l u_1 - u_2 + \sqrt{3} u_3 - \frac{bl}{d} u_4)$$

$$\Omega_6^2 = \frac{1}{6bl}(l u_1 + u_2 + \sqrt{3} u_3 + \frac{bl}{d} u_4)$$



$$\Omega_1^2 = \frac{1}{6bl}(l u_1 + 2 u_2 - \frac{bl}{d} u_4)$$

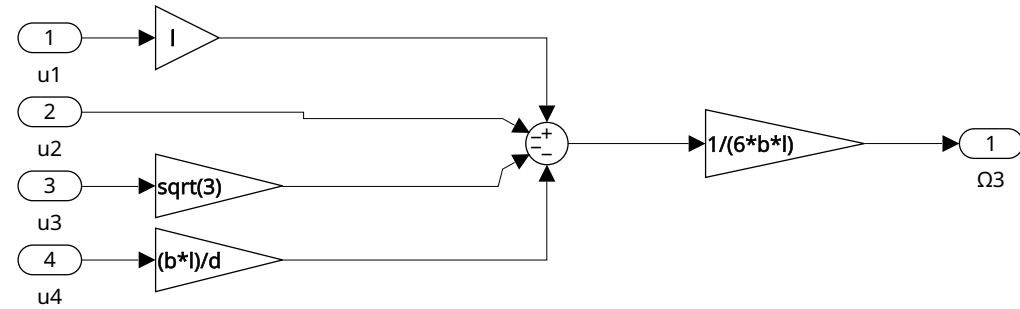
$$\Omega_2^2 = \frac{1}{6bl}(l u_1 + u_2 - \sqrt{3} u_3 + \frac{bl}{d} u_4)$$

$$\Omega_3^2 = \frac{1}{6bl}(l u_1 - u_2 - \sqrt{3} u_3 - \frac{bl}{d} u_4)$$

$$\Omega_4^2 = \frac{1}{6bl}(l u_1 - 2 u_2 + \frac{bl}{d} u_4)$$

$$\Omega_5^2 = \frac{1}{6bl}(l u_1 - u_2 + \sqrt{3} u_3 - \frac{bl}{d} u_4)$$

$$\Omega_6^2 = \frac{1}{6bl}(l u_1 + u_2 + \sqrt{3} u_3 + \frac{bl}{d} u_4)$$



$$\Omega_1^2 = \frac{1}{6bl}(l u_1 + 2 u_2 - \frac{bl}{d} u_4)$$

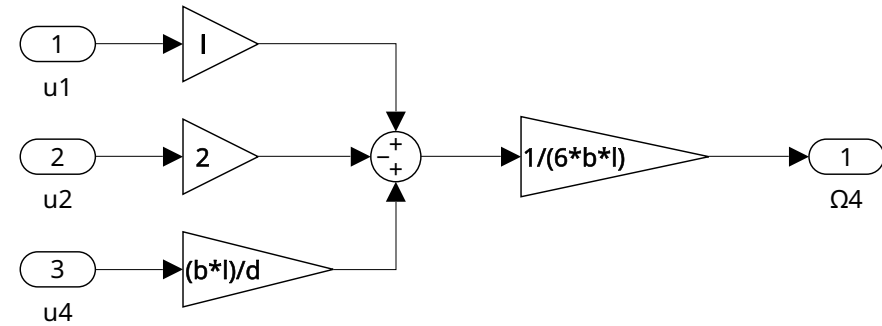
$$\Omega_2^2 = \frac{1}{6bl}(l u_1 + u_2 - \sqrt{3} u_3 + \frac{bl}{d} u_4)$$

$$\Omega_3^2 = \frac{1}{6bl}(l u_1 - u_2 - \sqrt{3} u_3 - \frac{bl}{d} u_4)$$

$$\Omega_4^2 = \frac{1}{6bl}(l u_1 - 2 u_2 + \frac{bl}{d} u_4)$$

$$\Omega_5^2 = \frac{1}{6bl}(l u_1 - u_2 + \sqrt{3} u_3 - \frac{bl}{d} u_4)$$

$$\Omega_6^2 = \frac{1}{6bl}(l u_1 + u_2 + \sqrt{3} u_3 + \frac{bl}{d} u_4)$$



$$\Omega_1^2 = \frac{1}{6bl}(l u_1 + 2 u_2 - \frac{bl}{d} u_4)$$

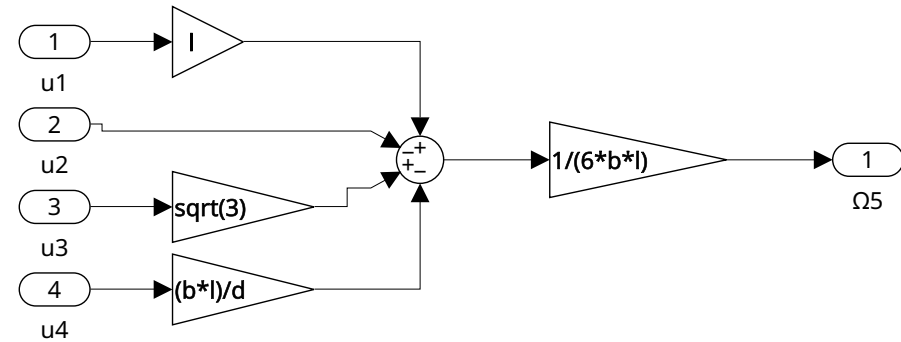
$$\Omega_2^2 = \frac{1}{6bl}(l u_1 + u_2 - \sqrt{3} u_3 + \frac{bl}{d} u_4)$$

$$\Omega_3^2 = \frac{1}{6bl}(l u_1 - u_2 - \sqrt{3} u_3 - \frac{bl}{d} u_4)$$

$$\Omega_4^2 = \frac{1}{6bl}(l u_1 - 2 u_2 + \frac{bl}{d} u_4)$$

$$\Omega_5^2 = \frac{1}{6bl}(l u_1 - u_2 + \sqrt{3} u_3 - \frac{bl}{d} u_4)$$

$$\Omega_6^2 = \frac{1}{6bl}(l u_1 + u_2 + \sqrt{3} u_3 + \frac{bl}{d} u_4)$$



$$\Omega_1^2 = \frac{1}{6bl}(l u_1 + 2 u_2 - \frac{bl}{d} u_4)$$

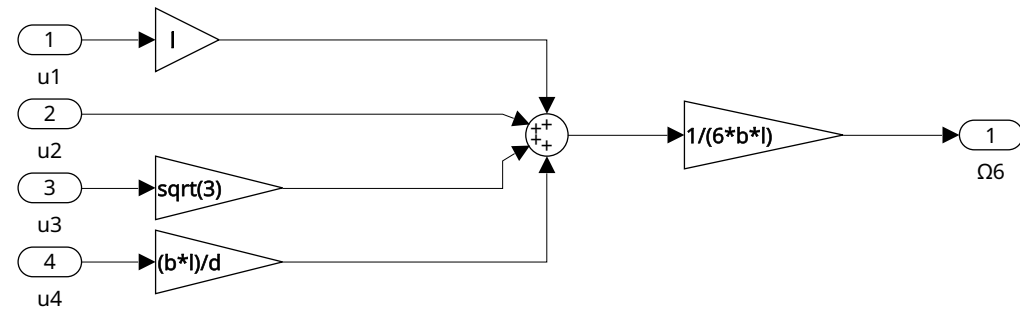
$$\Omega_2^2 = \frac{1}{6bl}(l u_1 + u_2 - \sqrt{3} u_3 + \frac{bl}{d} u_4)$$

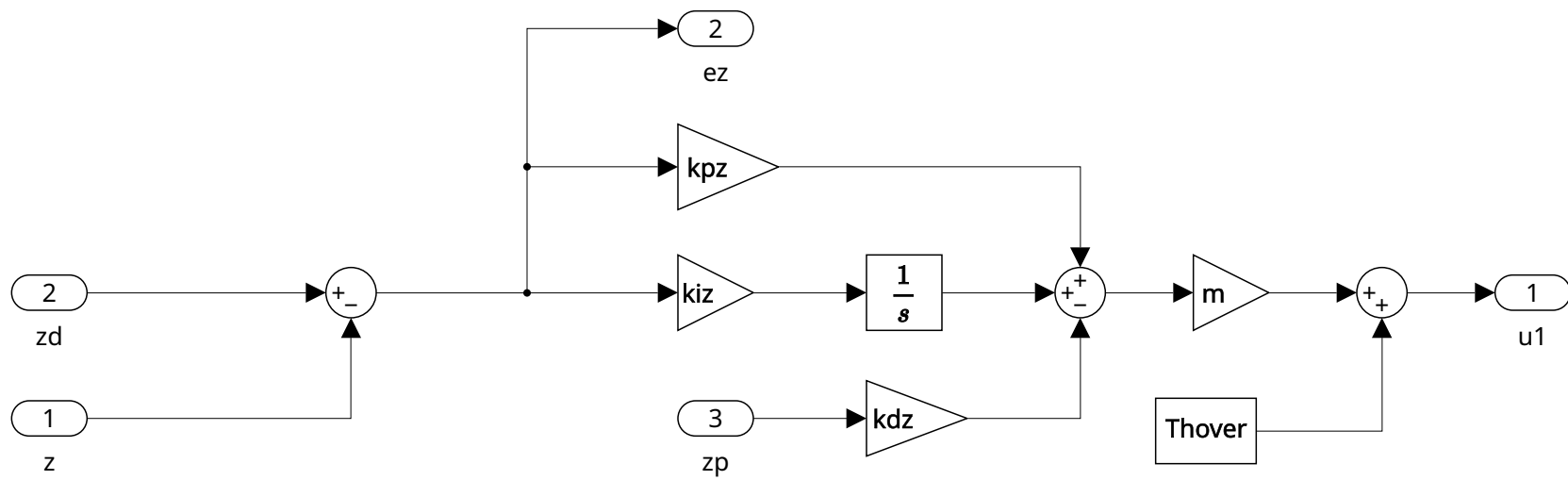
$$\Omega_3^2 = \frac{1}{6bl}(l u_1 - u_2 - \sqrt{3} u_3 - \frac{bl}{d} u_4)$$

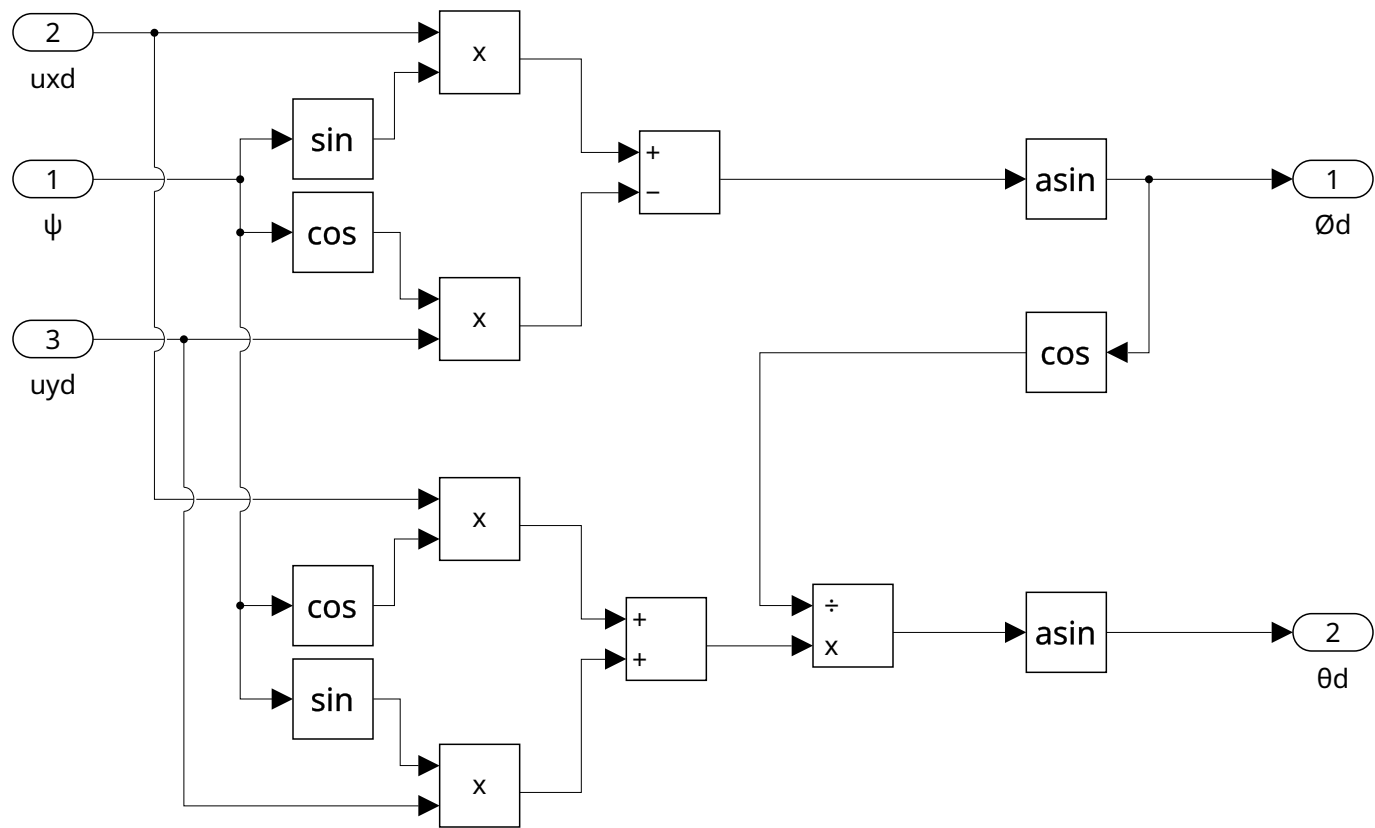
$$\Omega_4^2 = \frac{1}{6bl}(l u_1 - 2 u_2 + \frac{bl}{d} u_4)$$

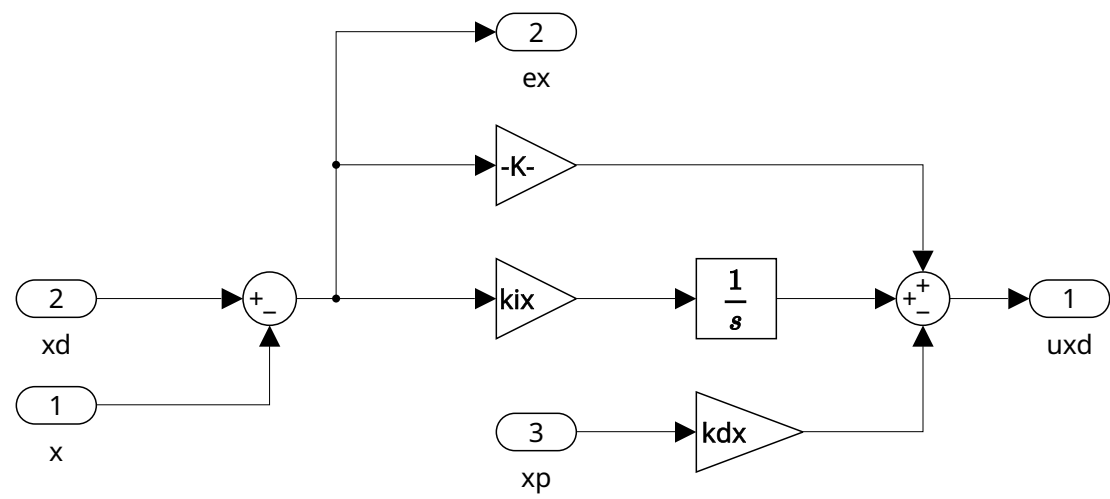
$$\Omega_5^2 = \frac{1}{6bl}(l u_1 - u_2 + \sqrt{3} u_3 - \frac{bl}{d} u_4)$$

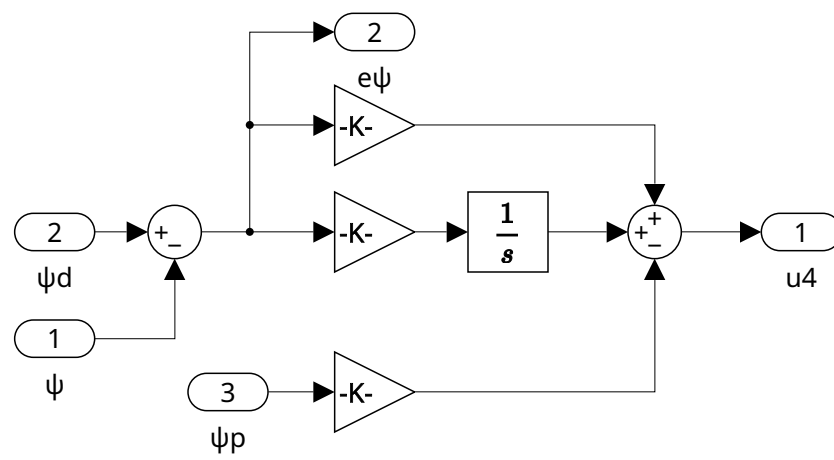
$$\Omega_6^2 = \frac{1}{6bl}(l u_1 + u_2 + \sqrt{3} u_3 + \frac{bl}{d} u_4)$$

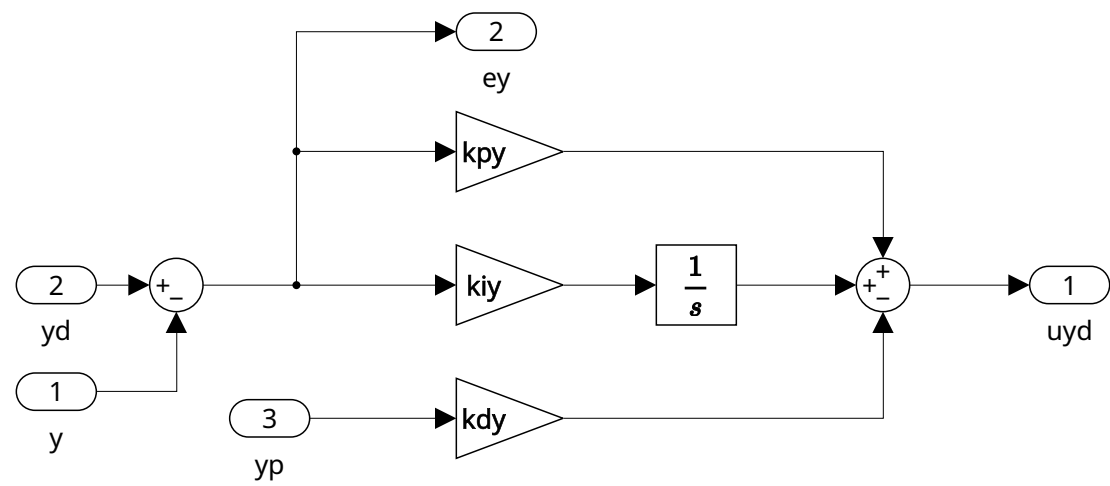






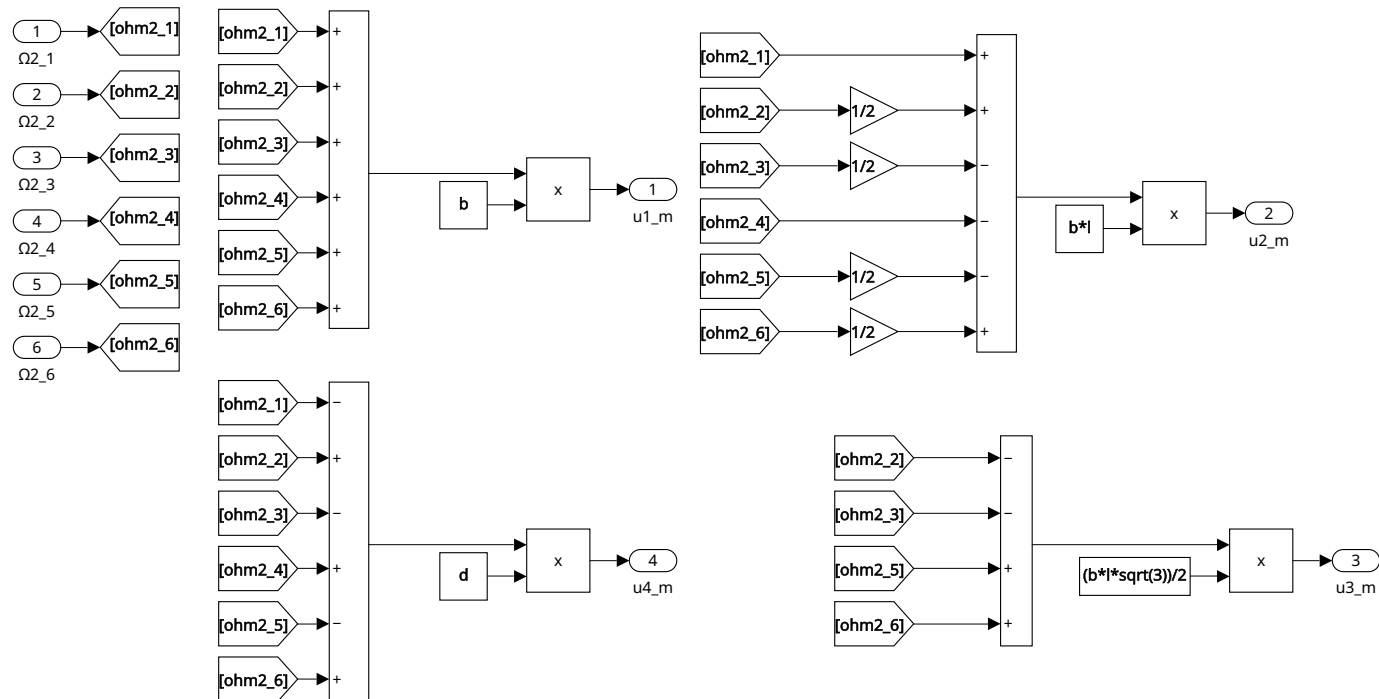






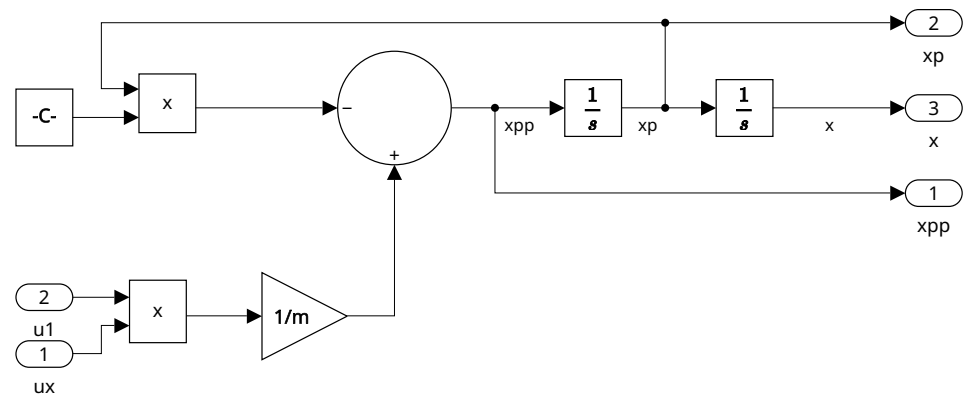
$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} b & b & b & b & b & b \\ \frac{-bl}{2} & -bl & \frac{-bl}{2} & \frac{bl}{2} & bl & \frac{bl}{2} \\ \frac{-bl\sqrt{3}}{2} & 0 & \frac{bl\sqrt{3}}{2} & \frac{bl\sqrt{3}}{2} & 0 & \frac{-bl\sqrt{3}}{2} \\ -d & d & -d & d & -d & d \end{bmatrix} \begin{bmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \\ \Omega_5^2 \\ \Omega_6^2 \end{bmatrix}$$

$$\begin{aligned} u_1 &= b (\Omega_1^2 + \Omega_2^2 + \Omega_3^2 + \Omega_4^2 + \Omega_5^2 + \Omega_6^2) \\ u_2 &= bl \left(\Omega_1^2 + \frac{\Omega_2^2}{2} - \frac{\Omega_3^2}{2} - \Omega_4^2 - \frac{\Omega_5^2}{2} + \frac{\Omega_6^2}{2} \right) \\ u_3 &= \frac{\sqrt{3}bl}{2} (-\Omega_2^2 - \Omega_3^2 + \Omega_5^2 + \Omega_6^2) \\ u_4 &= d (-\Omega_1^2 + \Omega_2^2 - \Omega_3^2 + \Omega_4^2 - \Omega_5^2 + \Omega_6^2) . \end{aligned}$$



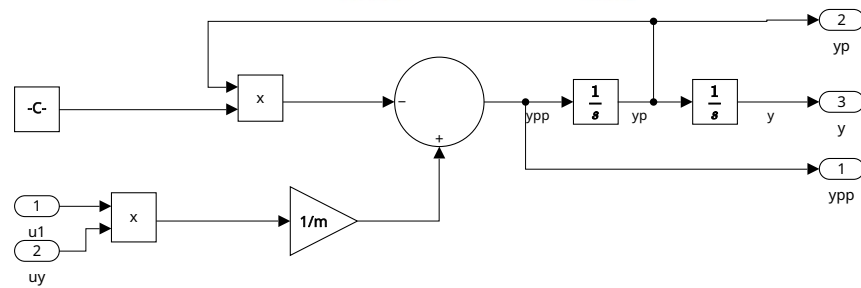
$$\ddot{x} = -\frac{k_f t_x}{m} \dot{x} + \frac{1}{m} u_x u_1$$

$$\ddot{x} = -\frac{k_f t_x}{m} \dot{x} + \frac{1}{m} u_x u_1$$

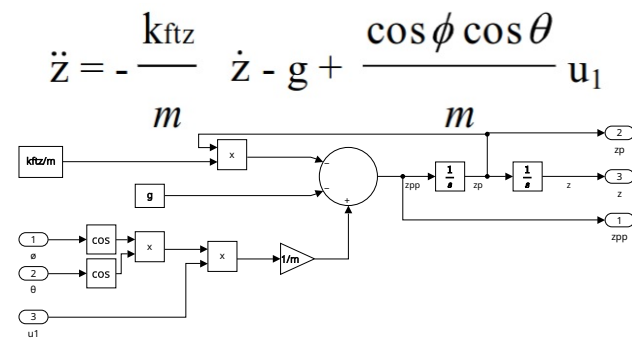


$$\ddot{y} = -\frac{k_{ft_y}}{m} \dot{y} + \frac{1}{m} u_y u_1$$

$$\ddot{y} = -\frac{k_{ft_y}}{m} \dot{y} + \frac{1}{m} u_y u_1$$



$$\ddot{z} = -\frac{k_{ftz}}{m} \dot{z} - g + \frac{\cos \phi \cos \theta}{m} u_1$$



$$\ddot{\psi} = \frac{1}{J_{zz}} [\dot{\phi} \dot{\theta} (J_{xx} - J_{zz}) - K_{faz} \dot{\psi}^2 + u_4]$$

$$\ddot{\psi} = \frac{1}{J_{zz}} [\dot{\phi} \dot{\theta} (J_{xx} - J_{zz}) - K_{faz} \dot{\psi}^2 + u_4]$$

